

Java Quick Reference

LIST (java.util.List)

```
List<String> x = new ArrayList<>(); // create an empty mutable list
List<String> x = List.of("a", "b"); // create an immutable list (can't be modified)
x.get(i) // returns element at index i
x.size() // returns number of elements
x.contains(item) // returns true if item is in the list
x.isEmpty() // returns true if list is empty
x.add(item) // appends item to end (mutable lists only)
```

MAP (java.util.Map)

```
Map<String, Integer> m = new HashMap<>(); // create an empty mutable map
Map<String, Integer> m = Map.of("a", 1); // create an immutable map (can't be modified)
m.put(key, value) // sets key to value (mutable maps only)
m.get(key) // returns value for key, or null if absent
m.getOrDefault(key, defaultValue) // returns value for key, or defaultValue if
absent
m.containsKey(key) // returns true if key is in the map
m.size() // returns number of entries
m.keySet() // returns a Set of all keys
```

SET (java.util.Set)

```
Set<String> s = new HashSet<>(); // create an empty mutable set
Set<String> s = Set.of("a", "b"); // create an immutable set (can't be modified)
s.add(item) // adds item (mutable sets only)
s.contains(item) // returns true if item is in the set
s.size() // returns number of elements
s.isEmpty() // returns true if set is empty
```

STRING (java.lang.String)

```
String s = "hello"; // string literal
s.length() // returns number of characters
s.charAt(i) // returns the character at index i
s.equals(other) // returns true if strings are equal (use instead of
==)
```

ITERATOR (java.util.Iterator)

```
it.hasNext() // returns true if there are more elements
it.next() // returns the next element
```

ITERABLE (java.lang.Iterable)

```
x.iterator() // returns an Iterator over the elements
for (T item : x) // for-each loop (works on any Iterable)
```

TESTING (com.google.common.truth.Truth)

```
assertThat(x).isEqualTo(y) // asserts that x equals y
```